Multiphoton interference and representation theory in the many-photon limit

PhD Candidacy Exam

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Introduction

Multiphoton interference in linear optical networks is a fundamental quantum effect and an important resource for quantum technologies. Recent experiments have used programmable single-photon processors to perform boson sampling and other quantum information tasks [1]. They have also shown how partial indistinguishability affects photon-number statistics and how collective photonic phases can change bunching probabilities [2, 3]. Large-scale Gaussian boson sampling experiments and advances in classical simulation have also highlighted the challenges of demonstrating quantum computational advantage [4, 5]. However, calculating multiphoton scattering amplitudes and probabilities becomes increasingly difficult as the number of photons increases, especially when partial indistinguishability is considered [6, 7]. This problem is closely related to boson sampling and computational complexity [8, 9, 10]. These difficulties motivate the search for an approximate description that can help us understand the behavior of multiphoton probabilities, even when their exact calculation is difficult.

Previous studies, including our own work, have established semiclassical correspondences between multiphoton interference and classical interferometry in the many-photon limit [11, 12]. These results provide one perspective on the problem, while representation theory provides another through the exact decomposition of the multiphoton mode space into irreducible sectors of $U(M)$. In this decomposition, sectors beyond the fully symmetric representation can also contribute when photons are partially distinguishable. Together, these results reveal connections between multiphoton interference, classical interferometry, and representation theory, while leaving open the question of how the contributions of the irreducible sectors are related to the classical description in the many-photon limit. These connections and the problems they raise form the starting point of this proposal.

This research focuses on the many-photon limit, $N \to \infty$, with a fixed number of modes M and sequences of occupations whose fractions approach fixed values. Figure 1 illustrates the three domains involved: quantum multiphoton interference, the classical map from relative phases to output intensities, and the theory of irreducible representations of $U(M)$, whose sectors are labeled by Young diagrams that grow with N.

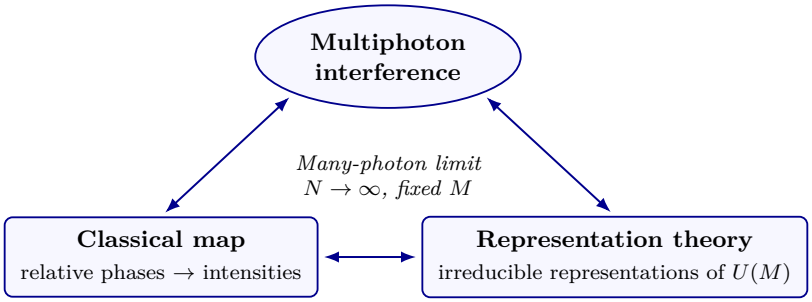


Figure 1: Correspondences between multiphoton interference, the classical map from relative phases to intensities, and the representation theory of $U(M)$ in the asymptotic multiphoton limit.

The first correspondence relates multiphoton probabilities to classical interferometry. In the many-photon limit, the macroscopic behavior of these probabilities is related to a classical measure induced by a map from relative phases to intensities [11, 12]. In our previous work, we expressed multiphoton states exactly through an expansion in tensor powers of auxiliary single-photon states, referred to as *virtual states* [12]. At zero relative phases, tracing over the internal degrees of freedom of a virtual state gives the effective single-photon matrix $\rho_{\overline{\boldsymbol{n}},\Gamma}$, where $\overline{\boldsymbol{n}} = \boldsymbol{n}/N$ denotes the normalized input occupations and Γ describes the internal-state overlaps. Varying the relative phases and propagating these virtual states through the interferometer defines the classical map $C_{\rho,U}(\boldsymbol{\phi})$, which maps the torus $\mathbb{T}^{M-1}$ to the simplex of normalized output intensities Δ_{M-1}. The probabilities are exponentially suppressed at a positive distance from the image of the map, and, at regular points, their rescaled slowly varying component converges to the density of the induced classical measure. Under additional conditions on the interferometer and the rank of the effective matrix, this convergence also holds for

the rescaled physical probabilities. When $M \geq 3$, the map may exhibit caustics and interior voids, the latter corresponding to regions without preimages. A general characterization of the interference contributions associated with different preimages, including their relative phases and behavior near caustics, as well as the determination of precise exponential suppression rates in classically forbidden regions, remains an open problem motivating this research.

The second correspondence relates multiphoton interference to representation theory. Schur–Weyl duality provides an exact decomposition of the multiphotonic state space into sectors $\mathcal{H}_\Lambda \otimes \mathcal{M}_\Lambda$, where $\mathcal{H}_\Lambda$ carries an irreducible representation of $U(M)$ labeled by a Young diagram and $\mathcal{M}_\Lambda$ is its Schur–Weyl multiplicity space [13, 14]. For two modes, previous work showed that the dominant spin sectors concentrate around a value that depends on the degree of indistinguishability [11]. For a fixed number of modes, the normalized Young diagrams of the relevant sectors concentrate around the spectrum of the effective single-photon matrix, as follows from Keyl–Werner bounds and conditioning on fixed input occupations [15]. However, this concentration does not determine the transition probabilities within each sector. In each sector, the interferometer acts through $D^\Lambda(U)$, the matrix representing the unitary transformation U in the irreducible space $\mathcal{H}_\Lambda$, while the physical preparation conditioned on a sector of nonzero probability is described by a density matrix $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$. For three or more modes, weight spaces may have more than one independent direction, and this density matrix is generally mixed and describes populations and possible coherences among these directions. The full multiphoton probability is an exact weighted sum of conditional sector probabilities, which depend on both $D^\Lambda(U)$ and $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$. Determining how the choice of internal states can favor particular irreducible sectors and affect populations and coherences within their weight spaces is a central problem of this research. Furthermore, extracting information about the representation matrix elements of $D^\Lambda(U)$ from the resulting multiphoton probabilities constitutes a key objective.

The third correspondence connects the classical map with the semiclassical behavior of irreducible representations of $U(M)$. In the fully symmetric case, our previous work [12] used expansions based on virtual single-photon states to study multiphoton amplitudes through properties of single-photon transition amplitudes. For other representations, we have constructed generalized coherent states in the three-mode case by acting with $D^\Lambda(V)$ on highest-weight states $|\Omega_\Lambda\rangle$. We refer to these states as *multicoherent states*. Their projections onto weight spaces $\mathcal{H}_{\Lambda,\boldsymbol{n}}$, when nonzero and normalized, define pure projected states. We call V a *compatible frame* when the mean occupation fractions of the transformed state, before projection, match the input occupation fractions. This compatibility condition is equivalent to stationarity, at zero relative phases, of the action associated with the norm of the projected state. These constructions allow us to define classical maps associated with irreducible representations and their compatible frames, $C_{\Lambda,V,U}(\boldsymbol{\phi})$, and establish exponential suppression bounds for transitions of projected states at a positive distance from the images of these maps, provided that their initial projected norms are positive and not exponentially small. Building on these results, we will develop semiclassical approximations for matrix elements of $D^\Lambda(U)$ beyond the fully symmetric case by identifying the contributions of stationary points, their prefactors, and the relative phases governing their interference.

Together, these three correspondences define the central problem of this doctoral proposal: to establish quantitative relations among multiphoton scattering probabilities with partial indistinguishability, the classical measure induced by the map from relative phases to intensities, and the contributions of irreducible sectors of $U(M)$ in the many-photon limit $N \to \infty$ with fixed M. In particular, we characterize how the conditional physical probabilities of the irreducible sectors contribute to the full multiphoton distribution and how their contributions relate to the classical description. To address this problem, we will evaluate interference near caustics and precise suppression rates in classically forbidden regions, including Airy- and WKB-type approximations where applicable, and determine how internal-state overlaps affect sector contributions and what information multiphoton probabilities can provide about representation matrix elements. Finally, we will develop semiclassical approximations for matrix elements of irreducible representations beyond the fully symmetric sector, using virtual states and constructions based on projected states to determine stationary-point contributions, their prefactors, and relative phases.

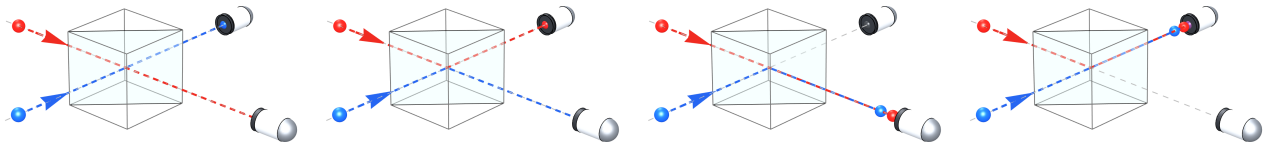
1 Physical motivation and the many-photon regime

In the physical setting considered here, N photons enter a passive, lossless linear interferometer with M optical modes and definite photon-number occupations at the input ports, corresponding to Fock-state preparations. The interferometer transforms the spatial modes, and the detectors measure the photon numbers at the output without resolving the internal states of the photons. Multiphoton interference arises because different propagation alternatives can lead to the same output occupation. When these alternatives are indistinguishable, their probability amplitudes add coherently, while partial distinguishability modifies the interference through the overlaps between the internal states of the photons.

In classical interferometry, by contrast, light is described by optical fields with amplitudes and relative phases, and interference is observed in the resulting output intensities. Optical coherent states provide a quantum description of such classical-like fields, but generally do not have definite photon numbers. When restricted to a fixed total photon number, these states can be expressed as symmetric tensor powers of single-photon states, although they do not generally have fixed occupations in each input mode. This contrast motivates the study of conditions under which the macroscopic behavior of multiphoton detection probabilities can be related to a classical description in the many-photon limit.

1.1 From two photons to interference in multiple modes

The Hong–Ou–Mandel effect shows how indistinguishability changes the detection probabilities of two photons [16]. Consider two photons entering different ports of a balanced beam splitter simultaneously. Each photon has an internal state, denoted by $|\chi_1\rangle$ and $|\chi_2\rangle$, which may describe properties such as polarization or temporal profile. When these states coincide, the photons are perfectly indistinguishable, while orthogonal internal states make them distinguishable. Fig. 2 shows the four alternatives involving transmission (T) and reflection (R), associated with their input ports.



(a) Both transmitted (TT); $(1,1)$. (b) Both reflected (RR); $(1,1)$. (c) Red T, blue R; $(0,2)$. (d) Red R, blue T; $(2,0)$.

Figure 2: Propagation alternatives for two indistinguishable photons in a balanced beam splitter. The letters T and R denote transmission and reflection, respectively. The alternatives TT and RR lead to the output occupation $(1,1)$, and their amplitudes interfere destructively, while TR and RT lead to the occupations $(0,2)$ and $(2,0)$. The colors indicate the paths associated with the input ports and do not represent distinguishable internal states.

To detect one photon at each output, both photons can be transmitted (TT) or reflected (RR). For perfectly indistinguishable photons, the coincidence amplitude is the coherent sum of these two contributions. For the single-photon transformation of a balanced beam splitter, we choose

$$U_{\mathrm{BS}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \mathcal{A}_{1,1} = U_{11}U_{22} + U_{12}U_{21} = 0. \tag{1.1}$$

The amplitudes TT and RR have equal magnitudes and opposite signs, so the coincidence amplitude vanishes through destructive interference. If the internal states are orthogonal, the two alternatives no longer interfere, and their probabilities add, giving a coincidence probability of $1/2$. The bunching of indistinguishable photons results from interference, not from an attractive interaction.

Even with fixed M, increasing N can increase the number of alternatives leading to the same output occupation. Their interference depends on the relative phases and on the input and output occupations.

For perfectly indistinguishable photons, transition amplitudes are expressed in terms of permanents of matrices constructed from U, with rows and columns repeated according to the occupations [17]. Their direct evaluation can become computationally costly as the number of photons N increases.

Permanents also play a central role in boson sampling, where the number of modes may grow with N [8]. In contrast, this proposal addresses the asymptotic limit $N \to \infty$ with a fixed number of modes M. Before formalizing this regime, the input preparation and detection model with partial distinguishability must be specified.

1.2 Preparation, partial distinguishability, and output probabilities

Consider a linear, passive, lossless interferometer with M ports, described at the single-photon level by a unitary matrix $U \in U(M)$. At the input, N photons are distributed according to the occupation $\boldsymbol{n} = (n_1, \ldots, n_M)$, with $\sum_i n_i = N$. The interferometer acts on the spatial modes, while the detectors record the output occupation $\boldsymbol{n}' = (n'_1, \ldots, n'_M)$, with $\sum_i n'_i = N$, without resolving internal degrees of freedom such as polarization, frequency, or temporal profile.

The analysis is restricted to the family of preparations considered in our previous work: the n_i photons entering port i share a pure, normalized internal state $|\chi_i\rangle$, while states associated with different ports may partially overlap. This assumption excludes arbitrary internal preparations, such as those with varying states for photons entering the same port or correlations not described by a common state for each port. Within this family, distinguishability between ports is characterized by the Gram matrix

$$\Gamma_{ij} = \langle \chi_j | \chi_i \rangle, \qquad \Gamma \succeq 0, \qquad \Gamma_{ii} = 1. \tag{1.2}$$

When all internal states coincide, $\Gamma_{ij} = 1$ and the photons are perfectly indistinguishable. At the opposite extreme, $\Gamma = I$ corresponds to orthogonal internal states for photons entering different ports, although photons entering the same port remain indistinguishable from one another. For partial overlaps, the complex entries of Γ determine which contributions can interfere coherently and how they combine in the detection probabilities [6, 7].

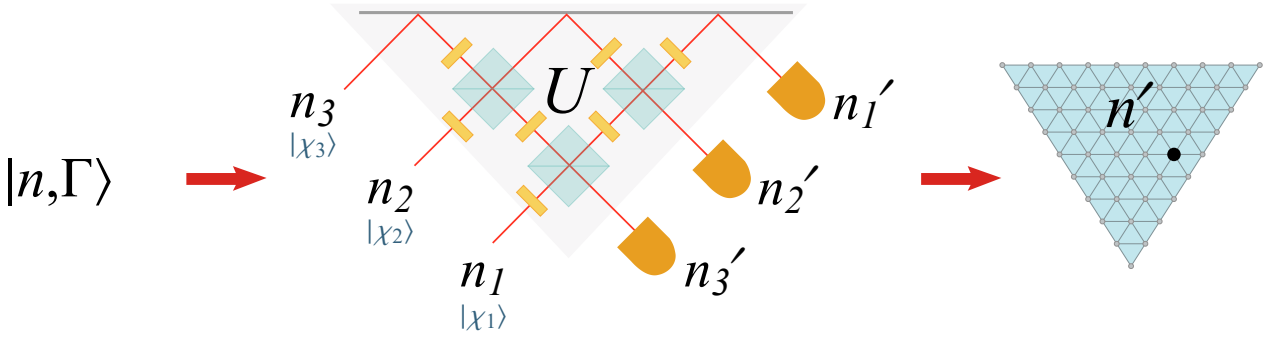


Figure 3: Schematic of the family of preparations considered in this work for a three-port interferometer. The n_i photons entering port i share an internal state $|\chi_i\rangle$, so partial distinguishability between ports is encoded in the overlaps $\Gamma_{ij} = \langle \chi_j | \chi_i \rangle$. The interferometer, described by the unitary matrix U, acts only on the spatial modes, while the detectors record the output occupations without resolving the internal states. Figure adapted from Fig. 1 of [12].

As illustrated in Fig. 3, for the same input occupation and interferometer U, changing the internal states $|\chi_i\rangle$ changes the overlaps in Γ and may change the output probabilities. The probability of detecting the occupation $\boldsymbol{n}'$ is denoted by $P(\boldsymbol{n}' \mid U, \boldsymbol{n}, \Gamma)$. Although exact expressions are available, the presence of spatial and internal degrees of freedom makes it difficult to identify interference patterns as the number of photons increases [6, 7].

To characterize the macroscopic behavior of these probabilities, we consider the limit $N \to \infty$ with M, U, and Γ fixed, and sequences of input and output occupations whose fractions approach fixed values:

$$N \longrightarrow \infty, \qquad M \text{ fixed}, \qquad \frac{\boldsymbol{n}}{N} \longrightarrow \overline{\boldsymbol{n}}, \qquad \frac{\boldsymbol{n}'}{N} \longrightarrow \overline{\boldsymbol{n}}'. \tag{1.3}$$

The normalized output occupations belong to the probability simplex $\Delta_{M-1} = \{\overline{\boldsymbol{n}}' : \overline{n}'_i \geq 0,\ \sum_i \overline{n}'_i = 1\}$. For $M = 3$, the simplex is a triangle whose points represent the fractions of photons detected at each port. For finite N, the possible occupations form a discrete lattice on the simplex that becomes finer as N increases.

The objective is to understand the macroscopic behavior of the output probabilities in this limit. In the next chapter, the input occupations and the Gram matrix Γ are used to define an effective single-photon matrix and the corresponding classical map.

2 Semiclassical structure of multiphoton interference

In the many-photon limit, multiphoton scattering probabilities can be studied through a classical map from relative input phases to output intensities [11, 12]. This map allows us to identify regions of exponential suppression and provides a starting point for studying the behavior of probabilities near caustics and the exact suppression rates.

2.1 Effective single-photon matrix and relative phases

For the preparations of Sec. 1.2, let $\overline{\boldsymbol{n}} = \boldsymbol{n}/N$ denote the normalized input occupations. In our previous work, we introduced auxiliary single-photon states, called virtual states, to study multiphoton interference [12]. At zero relative phases, one such state is

$$|\psi\rangle = \sum_{i=1}^{M} \sqrt{\overline{n}_i}\, |i\rangle \otimes |\chi_i\rangle,$$

where $|i\rangle$ denotes the input mode and $|\chi_i\rangle$ its associated internal state. The density matrix of this virtual state is $|\psi\rangle\langle\psi|$. Taking the partial trace over the internal degrees of freedom gives the effective single-photon matrix,

$$\rho_{\overline{\boldsymbol{n}},\Gamma} = \mathrm{Tr}_{\mathrm{int}}(|\psi\rangle\langle\psi|) = [\overline{\boldsymbol{n}}]^{1/2}\Gamma[\overline{\boldsymbol{n}}]^{1/2}, \qquad [\overline{\boldsymbol{n}}] = \mathrm{diag}(\overline{n}_1, \ldots, \overline{n}_M). \tag{2.1}$$

Since $\Gamma \succeq 0$ and $\Gamma_{ii} = 1$, this matrix is positive semidefinite and satisfies

$$\mathrm{diag}\, \rho_{\overline{\boldsymbol{n}},\Gamma} = \overline{\boldsymbol{n}}, \qquad \mathrm{Tr}\, \rho_{\overline{\boldsymbol{n}},\Gamma} = 1. \tag{2.2}$$

The diagonal elements of $\rho_{\overline{\boldsymbol{n}},\Gamma}$ give the normalized input occupations, while its off-diagonal elements contain the internal-state overlaps weighted by these occupations. In the classical description, this matrix describes the input intensities and field correlations.

To introduce relative phases, we define

$$t_{\boldsymbol{\phi}} = \mathrm{diag}\left(e^{\mathrm{i}\phi_1}, \ldots, e^{\mathrm{i}\phi_M}\right). \tag{2.3}$$

Removing the global phase by setting $\phi_1 = 0$ leaves $M - 1$ independent phases forming the torus $\mathbb{T}^{M-1}$. The corresponding virtual states are

$$|\psi_{\boldsymbol{\phi}}\rangle = (t_{\boldsymbol{\phi}} \otimes I)|\psi\rangle = \sum_{i=1}^{M} \sqrt{\overline{n}_i}\, e^{\mathrm{i}\phi_i} |i\rangle \otimes |\chi_i\rangle.$$

The fixed-occupation multiphoton state can then be expressed exactly as a coherent superposition of tensor powers of these virtual states:

$$|\boldsymbol{n},\Gamma\rangle = \frac{1}{\sqrt{\mathcal{M}(\boldsymbol{n}\mid\overline{\boldsymbol{n}})}}\int_{\mathbb{T}^{M-1}} d\mu(\boldsymbol{\phi})\, e^{-\mathrm{i}\boldsymbol{n}\cdot\boldsymbol{\phi}}|\psi_{\boldsymbol{\phi}}\rangle^{\otimes N}.$$

Here $d\mu(\boldsymbol{\phi})$ is the normalized uniform measure on the torus, and $\mathcal{M}(\boldsymbol{n}\mid\overline{\boldsymbol{n}})$ is the multinomial normalization factor. The virtual states are auxiliary states used to reconstruct the physical fixed-occupation preparation. Their propagation through U leads to an exact double-integral representation of the output probabilities (Propositions 1 and 2 of Ref. [12]) and provides the starting point for the classical map introduced in the next section.

2.2 Classical map, allowed region, and induced measure

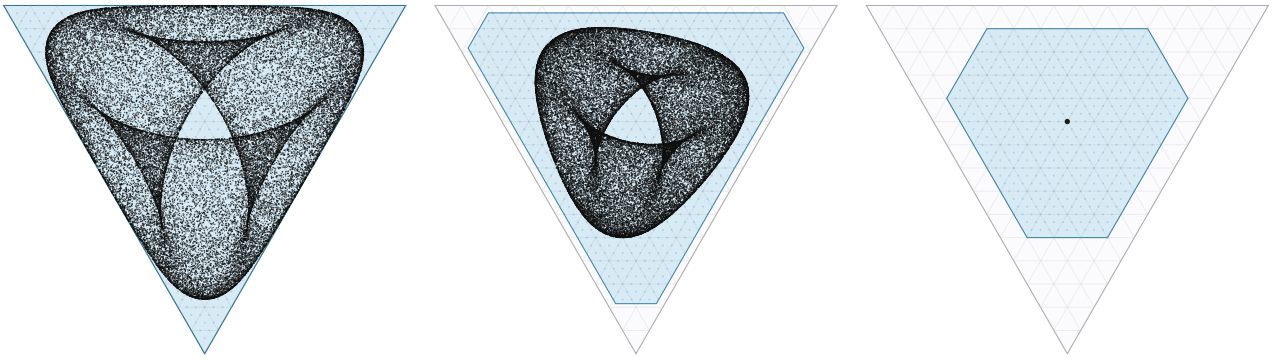
The propagation of the effective matrix through U, modulated by the phases $\boldsymbol{\phi}$, defines the classical map

$$C_{\rho,U}(\boldsymbol{\phi}) = \mathrm{diag}\left(U t_{\boldsymbol{\phi}} \rho_{\overline{\boldsymbol{n}},\Gamma} t_{\boldsymbol{\phi}}^{\dagger} U^{\dagger}\right), \qquad C_{\rho,U} : \mathbb{T}^{M-1} \longrightarrow \Delta_{M-1}, \tag{2.4}$$

where Δ_{M-1} is the simplex of normalized intensities. The image of the map is the classically allowed region,

$$\mathcal{R}_{\rho,U} = C_{\rho,U}(\mathbb{T}^{M-1}) \subseteq \Delta_{M-1}. \tag{2.5}$$

For $M = 3$, Fig. 4 compares the images of the map on the triangular simplex for three degrees of indistinguishability, with the input occupation and interferometer fixed. Here F_3 denotes the three-mode discrete Fourier transform. By the Schur–Horn theorem, these images are contained in the corresponding spectral permutahedra, shown in blue, but need not coincide with them.



(a) Perfect indistinguishability. (b) Partial distinguishability. (c) Orthogonal internal states between ports.

Figure 4: Images of the classical map for $N = 45$ photons, with input occupation $\boldsymbol{n} = (30, 12, 3)$ and $U = F_3$, for three degrees of indistinguishability: (a) perfectly indistinguishable photons, $\Gamma_{ij} = 1$; (b) partial indistinguishability, $\Gamma = \Gamma_{\mathrm{par}}$, with $(\Gamma_{\mathrm{par}})_{12} = (\Gamma_{\mathrm{par}})_{13} = 1/\sqrt{2}$ and $(\Gamma_{\mathrm{par}})_{23} = (3 - \mathrm{i}\sqrt{3})/8$, together with unit diagonal entries and conjugate off-diagonal entries; and (c) orthogonal internal states between ports, $\Gamma = I_3$. The black points are images of relative phases sampled on the torus, while the blue regions represent the corresponding spectral permutahedra.

The induced classical measure is obtained by pushing forward the normalized uniform measure on the torus to the simplex through $C_{\rho,U}$. At regular points $\overline{\boldsymbol{n}}'$, with discrete preimages and a nonzero reduced Jacobian, its density with respect to the independent simplex coordinates is

$$f_{\mathrm{clas}}(\overline{\boldsymbol{n}}' \mid U, \overline{\boldsymbol{n}}, \Gamma) = \frac{\mathbf{1}_{\mathcal{R}_{\rho,U}}(\overline{\boldsymbol{n}}')}{(2\pi)^{M-1}} \sum_{\boldsymbol{\phi}_\alpha :\, C_{\rho,U}(\boldsymbol{\phi}_\alpha)=\overline{\boldsymbol{n}}'} \left|\det\left(\frac{\partial C_{\rho,U}}{\partial \boldsymbol{\phi}}\right)_{\boldsymbol{\phi}_\alpha}\right|^{-1}. \tag{2.6}$$

The determinant is the reduced Jacobian in independent coordinates (for example, $\phi_1 = 0$ and components $C_2, \ldots, C_M$); each preimage contributes the inverse of its absolute value. Caustics may

arise where this Jacobian vanishes, at critical points where the regular formula fails and a different asymptotic treatment is needed. For $M \geq 3$, interior caustics and voids (interior regions without preimages) may occur. If $\Gamma = I_3$ and $U = F_3$, the image is the single point $(1/3, 1/3, 1/3)$, and the measure is concentrated at that point. The map delimits the forbidden region but does not, by itself, determine the quantum probabilities there or their behavior near caustics; the following theorems specify the scope of this correspondence [12].

2.3 Established asymptotic results

Three theorems of Ref. [12] make this correspondence precise.

Concentration in the classical region. Theorem 3 establishes the bound

$$P(\boldsymbol{n}' \mid U, \boldsymbol{n}, \Gamma) \leq (N+1)^M \exp\!\left[-N D(\overline{\boldsymbol{n}}' \| \mathcal{R}_{\rho,U})\right], \tag{2.7}$$

where $D(\overline{\boldsymbol{n}}' \| \mathcal{R}_{\rho,U}) = \min_{\boldsymbol{\nu} \in \mathcal{R}_{\rho,U}} D(\overline{\boldsymbol{n}}' \| \boldsymbol{\nu})$ and $D(\boldsymbol{x} \| \boldsymbol{y}) = \sum_i x_i \ln(x_i/y_i)$ is the relative entropy. At a positive distance from the allowed region, the bound implies exponential suppression as $N \to \infty$, but it determines neither the probabilities inside the region nor the exact suppression rates outside it. Its mathematical form is analogous to large-deviation bounds [18, 19].

Limit of the macroscopic envelope. P_{slow} collects the diagonal contributions of the double integral. At regular points, Theorem 4 shows that its rescaled density, $f_{\mathrm{slow}} = N^{M-1} P_{\mathrm{slow}}$, converges to the classical density:

$$\lim_{N\to\infty} f_{\mathrm{slow}}(\overline{\boldsymbol{n}}' \mid U, \overline{\boldsymbol{n}}, \Gamma) = f_{\mathrm{clas}}(\overline{\boldsymbol{n}}' \mid U, \overline{\boldsymbol{n}}, \Gamma). \tag{2.8}$$

This limit does not, by itself, imply that quantum oscillations disappear.

Suppression of rapid interference oscillations. For generic interferometers and full-rank $\rho_{\overline{\boldsymbol{n}},\Gamma}$, Theorem 5 establishes the asymptotic suppression of the off-diagonal interference contributions, so that

$$\lim_{N\to\infty} N^{M-1} P(\boldsymbol{n}' \mid U, \boldsymbol{n}, \Gamma) = f_{\mathrm{clas}}(\overline{\boldsymbol{n}}' \mid U, \overline{\boldsymbol{n}}, \Gamma), \tag{2.9}$$

both at regular points of the allowed region and in the forbidden region. This limit now concerns the rescaled physical probabilities.

If the initial occupation fractions are strictly positive, $\rho_{\overline{\boldsymbol{n}},\Gamma}$ and Γ have the same rank. Rank deficiency of Γ is necessary, but not sufficient, for rapid interference oscillations to persist asymptotically; perfect indistinguishability ($\operatorname{rank}\Gamma = 1$) provides an example. These theorems do not provide uniform approximations near caustics or a general characterization of persistent interference.

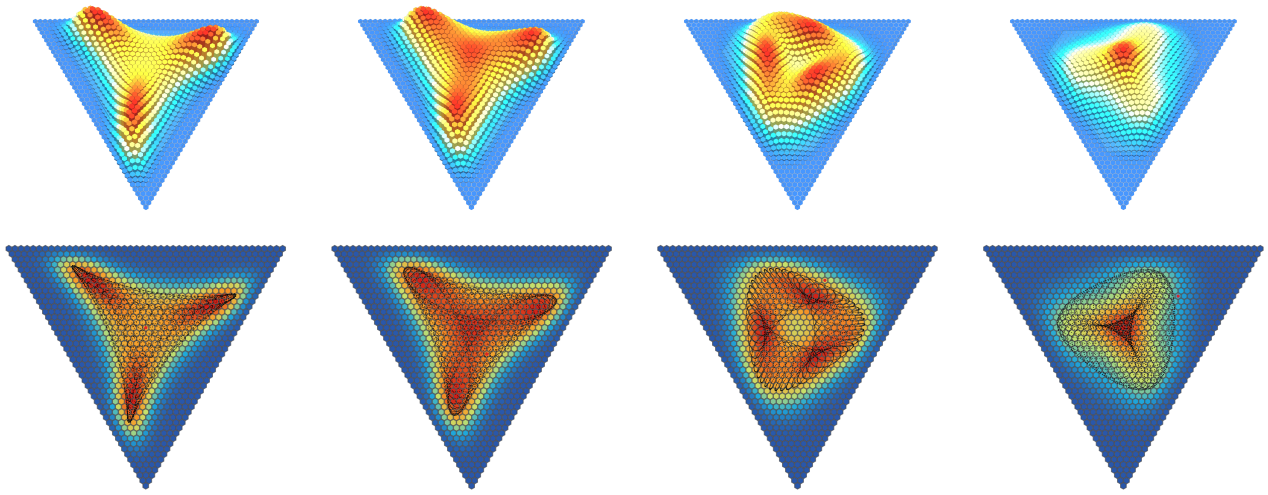
2.4 Comparison with quantum probabilities and scope of the correspondence

To compare multiphoton probabilities with the images of the classical map, four preparations are considered with $N = 45$, $U = F_3$, and a common effective spectrum $\boldsymbol{\lambda} = (11, 3, 1)/15$ (Table 1).

Table 1: Input occupations of the four preparations considered in Fig. 5, for $N = 45$ photons. The occupations $\boldsymbol{n}$ and their normalized forms $\overline{\boldsymbol{n}} = \boldsymbol{n}/N$ are shown.

Panel	$\boldsymbol{n}$	$\overline{\boldsymbol{n}}$
(a)	$(15, 15, 15)$	$(1, 1, 1)/3$
(b)	$(15, 10, 20)$	$(3, 2, 4)/9$
(c)	$(24, 15, 6)$	$(8, 5, 2)/15$
(d)	$(27, 9, 9)$	$(3, 1, 1)/5$

By the Schur–Horn theorem, $\overline{\boldsymbol{n}}^{(j)} \prec \boldsymbol{\lambda}$ guarantees the existence of effective matrices with that spectrum and diagonal, without uniquely determining their internal-state overlaps. Fig. 5 compares the output probabilities with sampled images of the map on the simplex.



(a) Balanced $\boldsymbol{n} = (15, 15, 15)$. (b) Asymmetric $\boldsymbol{n} = (15, 10, 20)$. (c) Intermediate $\boldsymbol{n} = (24, 15, 6)$. (d) More pronounced $\boldsymbol{n} = (27, 9, 9)$.

Figure 5: Comparison between multiphoton detection probabilities and the classical map for four preparations of $N = 45$ photons, with $U = F_3$ and a common effective spectrum $\boldsymbol{\lambda} = (11, 3, 1)/15$. The upper panels show the output probabilities using heights and colors, while the lower panels show the same probabilities on the simplex of normalized occupations, together with the black points obtained by sampling the classical map. The input occupations are indicated below each column.

For $\boldsymbol{\lambda} = \operatorname{spec} \rho_{\overline{\boldsymbol{n}},\Gamma}$, the Schur–Horn theorem implies

$$\mathcal{R}_{\rho,U} \subseteq \Pi(\boldsymbol{\lambda}), \qquad \Pi(\boldsymbol{\lambda}) = \operatorname{conv}\{\pi\boldsymbol{\lambda} : \pi \in S_M\}. \tag{2.10}$$

The spectral permutahedron $\Pi(\boldsymbol{\lambda})$ does not necessarily coincide with the image of the map [12, 20]. With the same spectrum and interferometer, the four preparations give different images and quantum distributions whose macroscopic patterns reflect these differences. Thus, the spectrum alone determines neither the image of the map nor the output probabilities.

The comparison for $N = 45$ illustrates, but does not prove, the asymptotic limits of Sec. 2.3. At regular points, the induced classical measure describes the macroscopic envelope, but does not determine the individual contributions of the irreducible sectors of $U(M)$ or semiclassical approximations for matrix elements beyond the fully symmetric representation. These connections provide the basis for the following chapters.

3 Multiphoton interference and representation theory

Chapter 2 related multiphoton probabilities to the measure induced by the classical map [12]. Starting from this macroscopic description, we present the exact decomposition into irreducible sectors of $U(M)$, whose internal structure allows us to determine conditional probabilities that the classical map itself does not specify.

3.1 Input configuration, interference, and irreducible sectors

Figure 6 illustrates the three correspondences underlying this research through a common three-mode configuration. We consider $N = 45$ photons, initial occupation $\boldsymbol{n} = (15, 10, 20)$, and interferometer $U = F_3$. The irreducible polynomial representation of $GL(3, \mathbb{C})$ labeled by the partition $\Lambda = (33, 9, 3)$ is considered. Its restriction to $U(3)$ remains irreducible, and its further restriction to $SU(3)$ has Dynkin labels $(p, q) = (24, 6)$. We also choose a unitary matrix $V_1 \in U(3)$ satisfying $\operatorname{diag}(V_1[\overline{\Lambda}]V_1^\dagger) = \overline{\boldsymbol{n}}$, where $\overline{\Lambda} = \Lambda/N$. This choice determines an effective single-photon matrix with the prescribed spectrum and

diagonal, and an associated Gram matrix $\Gamma(V_1)$. We refer to unitary matrices satisfying this condition as *compatible frames*; their construction is developed in the following chapter. The three panels show the total physical multiphoton detection probabilities, the image of the corresponding classical map, and the three-dimensional distribution of transition intensities obtained from the projected pure state associated with V_1 in the selected irreducible representation Λ.

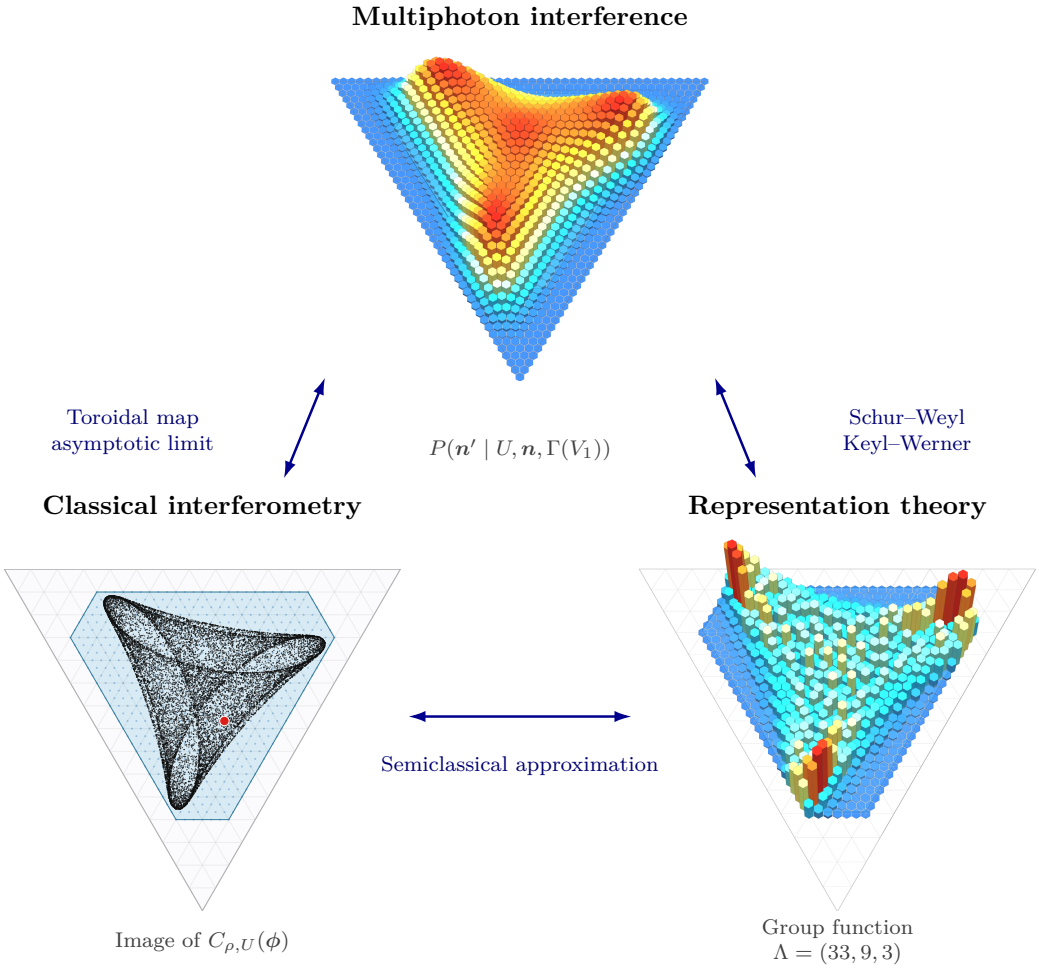


Figure 6: Correspondences between multiphoton interference, classical interferometry, and representation theory for $M = 3$, $N = 45$, $\boldsymbol{n} = (15, 10, 20)$, $U = F_3$, and the selected compatible frame V_1. Top: physical multiphoton detection probabilities, including contributions from all irreducible sectors. Left: sampled image of the classical map (black), spectral permutahedron (blue), and normalized input occupation $\overline{\boldsymbol{n}} = (1/3, 2/9, 4/9)$ (red). Right: three-dimensional distribution of group-function intensities for the projected pure state associated with V_1 in $\Lambda = (33, 9, 3)$, obtained by summing the squared moduli of the outgoing isospin amplitudes. The upper connections indicate the quantum–classical asymptotic correspondence and the Schur–Weyl decomposition with Keyl–Werner concentration. The lower connection indicates the proposed semiclassical approximation of representation matrix elements beyond the fully symmetric case.

The first correspondence relates physical multiphoton probabilities to the classical map. As established in Chapter 2, the exact toroidal expansion yields a classical measure induced by the map from relative phases to normalized output intensities [12]. The black points in the lower-left panel sample the image of the torus under the classical map, which is contained in the spectral permutahedron but need not coincide with it. In the many-photon limit, the rescaled slowly varying component of the probabilities converges to the induced classical density at regular points. Under additional conditions on the interferometer and the rank of the effective matrix, this also holds for the rescaled physical probabilities. Even at the moderate photon number $N = 45$, the figure already shows how the structure

of the physical multiphoton distribution is related to the image of the classical map.

The second correspondence relates multiphoton interference to representation theory through the exact Schur–Weyl decomposition, which partitions physical probabilities into irreducible sectors of $GL(3,\mathbb{C})$ labeled by Young diagrams. In the many-photon limit, the normalized Young diagrams concentrate around the effective single-photon spectrum under the corresponding assumptions [15]. For the selected configuration, $\overline{\Lambda} = (11, 3, 1)/15$ coincides with this spectrum. The initial occupation $\boldsymbol{n} = (15, 10, 20)$ belongs to a weight space of dimension $m_\Lambda(\boldsymbol{n}) = \dim \mathcal{H}_{\Lambda,\boldsymbol{n}} = 7$, distinct from the Schur–Weyl multiplicity. Within each sector, the interferometer acts through $D^\Lambda(U)$, while the conditional physical probabilities also depend on the populations and coherences of $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$.

The third correspondence connects the classical map with the semiclassical behavior of representation matrix elements in irreducible representations of $U(3)$. The lower-right panel shows the transition intensities obtained from the projected pure state associated with V_1. These intensities are calculated from the matrix elements of $D^\Lambda(U)$ by combining the input isospin amplitudes and summing their squared moduli over the outgoing isospin channels. The lower-left panel shows the image of the classical map associated with the same frame V_1 and interferometer U. The central problem is to characterize how the geometry of this map, including its preimages and caustics, relates to the asymptotic behavior of these transitions. Beyond the fully symmetric case, determining the stationary-point contributions, their prefactors, and the relative phases governing their interference remains an open problem. The following chapter develops projected-state constructions and associated classical maps to investigate this relation, without identifying these pure-state intensities with the physical probabilities conditioned on the irreducible sector.

3.2 Irreducible sectors, spectrum, and weight diagrams

With partial distinguishability, the reduced spatial description includes sectors that are not fully symmetric, although the full state retains its bosonic symmetry. For the correlated-mode preparations considered, the internal states span a space of dimension at most M, so both single-photon Hilbert spaces are identified with $\mathbb{C}^M$. The full N-photon Fock space then decomposes as

$$\mathcal{F}^{(N)} = \mathrm{Sym}^N\left(\mathbb{C}^M_{\mathrm{port}} \otimes \mathbb{C}^M_{\mathrm{int}}\right) \simeq \bigoplus_{\substack{\Lambda \vdash N \\ \ell(\Lambda) \le M}} \mathcal{H}_\Lambda \otimes \mathcal{H}^{(\mathrm{int})}_\Lambda. \tag{3.1}$$

In this decomposition, two factors carry irreducible polynomial representations of $GL(M,\mathbb{C})$ associated with the same Young diagram Λ, for the port and internal degrees of freedom, respectively. After tracing over the unresolved internal degrees of freedom, the reduced spatial state acts on $(\mathbb{C}^M)^{\otimes N}$, whose Schur–Weyl decomposition is

$$(\mathbb{C}^M)^{\otimes N} \simeq \bigoplus_{\substack{\Lambda \vdash N \\ \ell(\Lambda) \le M}} \mathcal{H}_\Lambda \otimes \mathcal{M}_\Lambda. \tag{3.2}$$

Each partition $\Lambda = (\Lambda_1, \ldots, \Lambda_M)$ of N labels an irreducible representation of $U(M)$ acting on $\mathcal{H}_\Lambda$, with multiplicity space $\mathcal{M}_\Lambda$ carrying the corresponding representation of S_N [13, 14]. For $M = 3$, its $SU(3)$ Dynkin labels are $(p, q) = (\Lambda_1 - \Lambda_2, \Lambda_2 - \Lambda_3)$.

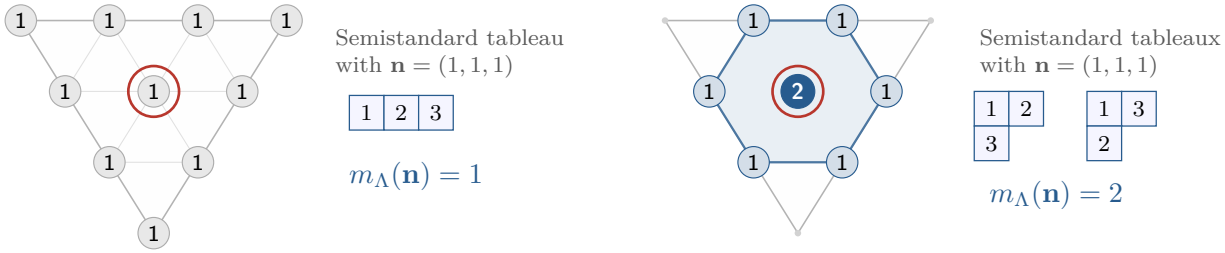
The sector weights are related to $\boldsymbol{\lambda} = \mathrm{spec}\,\rho_{\overline{\boldsymbol{n}},\Gamma}$. For product states $\rho^{\otimes N}$, Keyl and Werner proved the concentration of $\overline{\Lambda} = \Lambda/N$ around $\mathrm{spec}\,\rho$ [15]. When $\rho^{\otimes N}_{\overline{\boldsymbol{n}},\Gamma}$ is conditioned on the fixed occupation $\boldsymbol{n}$, whose probability is at least $(N+1)^{-M}$, the exponential bounds remain valid up to a polynomial factor (Appendix A). This concentration concerns the normalized irreducible labels, but does not determine the distribution within the weight spaces or the transition probabilities conditioned on each sector.

Each representation decomposes into weight spaces associated with the occupations:

$$\mathcal{H}_\Lambda = \bigoplus_{\boldsymbol{n}} \mathcal{H}_{\Lambda,\boldsymbol{n}}, \qquad m_\Lambda(\boldsymbol{n}) = \dim \mathcal{H}_{\Lambda,\boldsymbol{n}}. \tag{3.3}$$

The weight multiplicity $m_\Lambda(\boldsymbol{n})$ counts the internal directions associated with an occupation, distinct from the Schur–Weyl multiplicity $\dim \mathcal{M}_\Lambda$. The detectors do not resolve these directions, although the evolution may couple them.

The number of semistandard Young tableaux of shape Λ and content $\boldsymbol{n}$ is $m_\Lambda(\boldsymbol{n})$. For $M = N = 3$, Fig. 7 shows ten weights of multiplicity one for $\Lambda = (3,0,0)$ and seven for $\Lambda = (2,1,0)$, with multiplicity two at $\boldsymbol{n} = (1,1,1)$. The convex hull of the normalized weights is the permutahedron $\Pi(\overline{\Lambda})$, which generally differs from $\Pi(\boldsymbol{\lambda})$ and the classical image $\mathcal{R}_{\rho,U}$. These internal multiplicities occur for $U(3)$, but not in irreducible representations of $U(2)$.



(a) Symmetric: $\Lambda = (3,0,0)$. (b) Adjoint: $\Lambda = (2,1,0)$.

Figure 7: Weight diagrams and semistandard Young tableaux of two representations of $U(3)$ for $N = 3$: (a) fully symmetric representation, $\Lambda = (3,0,0)$; (b) representation $\Lambda = (2,1,0)$, corresponding to the adjoint of $SU(3)$. The numbers indicate the weight multiplicities. The marked weight, $\mathbf{n} = (1,1,1)$, has multiplicity one in (a) and two in (b), as shown by the corresponding tableaux.

3.3 From the symmetric case to multiplicity spaces

For perfectly indistinguishable photons, each occupation corresponds to a unique Fock state in $\mathcal{H}_{(N,0,\ldots,0)} = \mathrm{Sym}^N(\mathbb{C}^M)$; the amplitudes can be expressed in terms of permanents or representation matrix elements [17]. Semiclassical approximations are available for Wigner functions of $SU(2)$ and $SU(3)$ [21, 22]. In our previous work, known bosonic formulas were recovered using virtual single-photon states [12], motivating their extension to other representations.

With partial distinguishability, transition probabilities also depend on the preparation and the internal directions of each weight space.

The preparation is described by the normalized physical density matrix

$$\sigma_{\Lambda,\boldsymbol{n}}(\Gamma) = \frac{P_{\Lambda,\boldsymbol{n}} D^\Lambda(\Gamma) P_{\Lambda,\boldsymbol{n}}}{\mathrm{Tr}\left[P_{\Lambda,\boldsymbol{n}} D^\Lambda(\Gamma) P_{\Lambda,\boldsymbol{n}}\right]}. \tag{3.4}$$

Here $P_{\Lambda,\boldsymbol{n}}$ projects onto $\mathcal{H}_{\Lambda,\boldsymbol{n}}$, and $D^\Lambda(\Gamma)$ is the polynomial extension of the representation to Γ, which is not necessarily unitary. The density matrix is defined when $P(\Lambda \mid \boldsymbol{n}, \Gamma) > 0$ and is positive semidefinite with unit trace (Appendix A). In a weight-space basis, its diagonal and off-diagonal entries represent populations and coherences, respectively.

If Γ is positive definite, $\operatorname{rank} \sigma_{\Lambda,\boldsymbol{n}}(\Gamma) = m_\Lambda(\boldsymbol{n})$ (Appendix G); for $m_\Lambda(\boldsymbol{n}) > 1$, the physical density matrix is mixed and cannot be identified with a projected pure state. Determining the asymptotic role of its populations and coherences in the conditional physical probabilities of the irreducible sector remains an open problem.

3.4 Physical probability as a sum of sector contributions

The exact decomposition of the physical probability is (Appendices A and G):

$$\begin{aligned} P(\boldsymbol{n}' \mid U, \boldsymbol{n}, \Gamma) &= \sum_{\Lambda} P(\Lambda \mid \boldsymbol{n}, \Gamma)\, P(\boldsymbol{n}' \mid \boldsymbol{n}, \Lambda, U, \Gamma), \\ P(\boldsymbol{n}' \mid \boldsymbol{n}, \Lambda, U, \Gamma) &= \mathrm{Tr}\Big[P_{\Lambda,\boldsymbol{n}'} D^{\Lambda}(U) \sigma_{\Lambda,\boldsymbol{n}}(\Gamma) D^{\Lambda}(U)^{\dagger}\Big]. \end{aligned} \tag{3.5}$$

The first factor corresponds to the sector weight; the second, defined when $P(\Lambda \mid \boldsymbol{n}, \Gamma) > 0$, is the conditional probability. Only sectors with nonzero weight contribute to the sum; the trace accounts for internal directions not resolved by the detectors. The conditional probabilities depend on the matrix elements of $D^{\Lambda}(U)$ and on the populations and possible coherences of $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$. The identity is exact for finite N. Determining how internal-state overlaps affect the contributions of irreducible sectors and the populations and coherences within their weight spaces, and what information the resulting probabilities can yield about the representation matrix elements of $D^{\Lambda}(U)$, remains an open problem. The next chapter develops the compatible-frame condition, projected pure states, and classical maps associated with irreducible representations as tools for the semiclassical characterization of representation matrix elements, clearly distinguishing these pure states from the physical density matrix of a sector.

4 Progress toward semiclassical approximations for representation matrix elements

Chapter 3 highlighted the asymptotic behavior of transitions between weight spaces with internal multiplicity $m_{\Lambda}(\boldsymbol{n}) > 1$ as an open problem. For three modes, unitary matrices matching prescribed spectra and occupations, projected highest-weight states, and associated classical maps are employed to characterize the representation matrix elements of $D^{\Lambda}(U)$, distinguishing exact results, numerical evidence, and open questions.

4.1 Compatible frames and spectrum–diagonal geometry

Let $\Lambda = (\Lambda_1, \Lambda_2, \Lambda_3)$, $\sum_i \Lambda_i = N$, label an irreducible representation of $U(3)$, with $\overline{\Lambda} = \Lambda/N$. For $V \in U(3)$, consider the matrices

$$\rho_{\Lambda}(V) = V[\overline{\Lambda}]V^{\dagger}, \qquad Q(V)_{ij} = |V_{ij}|^2, \qquad \mathrm{diag}\,\rho_{\Lambda}(V) = Q(V)\overline{\Lambda}, \tag{4.1}$$

where $[\overline{\Lambda}]$ is diagonal and $Q(V)$ is unistochastic and therefore bistochastic.

In this work, we call a unitary matrix V a *compatible frame* for the normalized highest weight $\overline{\Lambda}$ and occupation $\overline{\boldsymbol{n}}$ if

$$Q(V)\overline{\Lambda} = \overline{\boldsymbol{n}}. \tag{4.2}$$

The Schur–Horn theorem guarantees the existence of compatible frames if and only if $\overline{\boldsymbol{n}} \prec \overline{\Lambda}$, without determining V or the off-diagonal elements of $\rho_{\Lambda}(V)$ [20, 23]. For three modes, the bistochastic matrices Q satisfying $Q\overline{\Lambda} = \overline{\boldsymbol{n}}$ are characterized, and unistochasticity is imposed to reconstruct the frames [24]; Heron's criterion for triangle closure is detailed in Appendix B [25, 26].

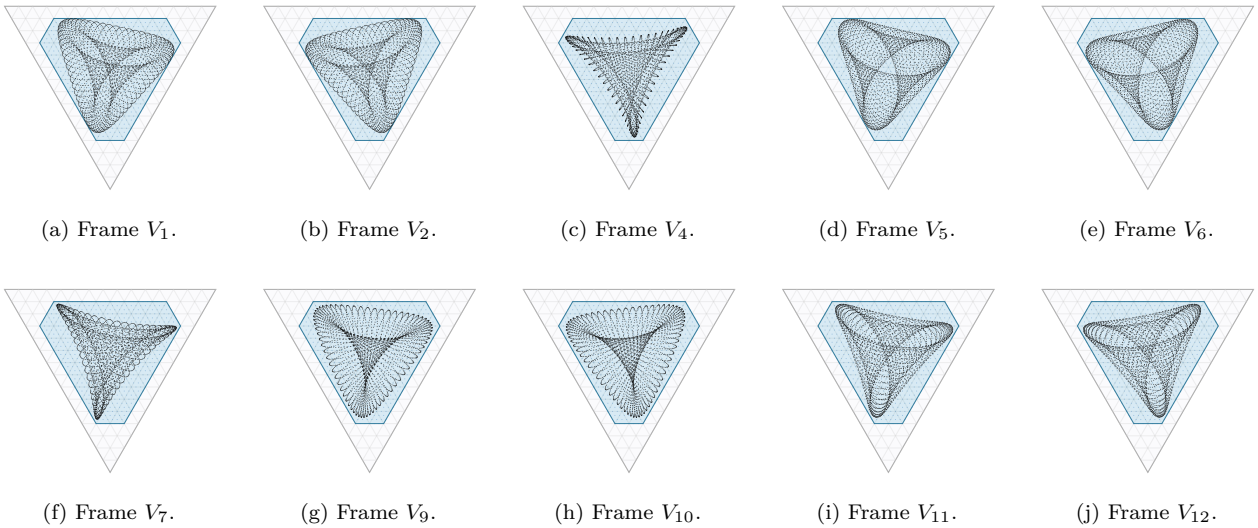
The matrix $\rho_{\Lambda}(V)$ associated with Λ and V need not coincide with the physical effective matrix $\rho_{\overline{\boldsymbol{n}},\Gamma}$. If $\overline{n}_i > 0$ for all i, each compatible frame defines an admissible Gram matrix,

$$\Gamma(V) = [\overline{\boldsymbol{n}}]^{-1/2} \rho_{\Lambda}(V) [\overline{\boldsymbol{n}}]^{-1/2}. \tag{4.3}$$

Different compatible frames may produce different matrices $\Gamma(V)$, even with the same spectrum and initial occupation. Agreement with the physical matrix requires $\overline{\Lambda} = \mathrm{spec}\,\rho_{\overline{\boldsymbol{n}},\Gamma}$ and that V diagonalizes

it. In that case, the classical maps coincide, but transition probabilities obtained from projected pure states need not agree with those determined by the physical density matrix within the sector.

Fig. 8 compares ten compatible frames with fixed Λ, $\overline{\boldsymbol{n}}$, and U. Although they share the same spectrum and diagonal, the images of their associated classical maps differ because they depend on V and the corresponding matrix $\Gamma(V)$. The points represent normalized classical intensities, not detection probabilities; the examples do not constitute an exhaustive classification.



(a) Frame V_1. (b) Frame V_2. (c) Frame V_4. (d) Frame V_5. (e) Frame V_6.

(f) Frame V_7. (g) Frame V_9. (h) Frame V_{10}. (i) Frame V_{11}. (j) Frame V_{12}.

Figure 8: Images of the classical map for ten compatible frames, with $N = 45$, $\Lambda = (33, 9, 3)$, $\overline{\boldsymbol{n}} = (1/3, 2/9, 4/9)$, and $U = F_3$. In each panel, the black points represent samples obtained by applying the map to relative phases on the torus, while the blue region corresponds to the common spectral permutahedron $\Pi(\overline{\Lambda})$. Although the spectrum, initial occupation, and interferometer remain fixed, the geometry of the image depends on the frame V.

4.2 Projected states and multiplicity directions in $SU(3)$

For perfectly indistinguishable photons, the relevant representation is fully symmetric, $\Lambda = (N, 0, 0)$. A normalized highest-weight vector is $|\Omega_\Lambda\rangle = |e_1\rangle^{\otimes N}$, whose orbit consists of the coherent states $D^\Lambda(V)|\Omega_\Lambda\rangle = |Ve_1\rangle^{\otimes N}$. Projection onto fixed-occupation weight spaces selects their Fock components and admits an exact Fourier representation over relative phases. This construction provides a link between representation matrix elements and the virtual-state method of Chapter 2, based on tensor powers of auxiliary single-photon states [12].

For general Λ, a normalized highest-weight vector $|\Omega_\Lambda\rangle \in \mathcal{H}_\Lambda$ is constructed in Appendix C from tensor products of exterior vectors $|e_1 \wedge \cdots \wedge e_r\rangle$ with multiplicities $p_r = \Lambda_r - \Lambda_{r+1}$. Acting with $D^\Lambda(V)$ generates its unitary orbit $D^\Lambda(V)|\Omega_\Lambda\rangle$, a family of generalized coherent states [27] that we call *multicoherent states* in this work. Projecting an orbit state onto the weight space $\mathcal{H}_{\Lambda,\boldsymbol{n}}$ and normalizing gives the pure state

$$|\boldsymbol{n}, V; \Lambda\rangle = \frac{P_{\Lambda,\boldsymbol{n}} D^\Lambda(V)|\Omega_\Lambda\rangle}{\sqrt{\mathcal{N}_{\Lambda,\boldsymbol{n}}(V)}}, \qquad \mathcal{N}_{\Lambda,\boldsymbol{n}}(V) = \left\| P_{\Lambda,\boldsymbol{n}} D^\Lambda(V)|\Omega_\Lambda\rangle \right\|^2, \tag{4.4}$$

provided that $\mathcal{N}_{\Lambda,\boldsymbol{n}}(V) > 0$. The projector onto a weight space has an exact Fourier representation over the torus of relative phases (Appendix C). In the fully symmetric representation, each weight space is one-dimensional. For more general representations, weight spaces may have dimension greater than one, and different choices of V can give distinct projected states at the same weight. A general semiclassical approximation for these transitions remains an open problem. For $SU(3)$, we use Gelfand–Tsetlin and isospin bases (Appendices C–F) [28, 29, 30].

In the adjoint representation of $SU(3)$, with $\Lambda = (2, 1, 0)$ and $\boldsymbol{n} = (1, 1, 1)$, two frames with the

same $Q(V)$ are constructed whose projected states are orthogonal (Appendix F). This exact result demonstrates the importance of the phases of V, without implying differences between their isospin populations.

The linear independence of projected states is examined through their Gram matrices. For $N = 45$, $\Lambda = (33, 9, 3)$, families of states are found numerically to span the seven-dimensional weight spaces corresponding to $\boldsymbol{n} = (15, 15, 15)$, $(27, 9, 9)$, and $(15, 10, 20)$ (Appendix F). The rank criterion is exact, but the observed spanning is based on numerical evidence and does not guarantee that every direction can be obtained from a single frame.

These pure projected states are not, in general, equivalent to the physical density matrix $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$ within the corresponding sector. Their transition probabilities are given by

$$P_{\text{frame}}(\boldsymbol{n}') = \left\| P_{\Lambda,\boldsymbol{n}'} D^{\Lambda}(U) |\boldsymbol{n}, V; \Lambda\rangle \right\|^2. \tag{4.5}$$

The conditional physical probabilities follow Eq. (3.5); for $N = 45$, the two descriptions are compared numerically and the isospin distributions are analyzed. By varying the compatible frame, the projected states can probe different directions of a weight space, as illustrated by the exact and numerical examples above. Whether this construction provides a general parametrization of those directions, and how it relates to physical density matrices, remain open questions.

4.3 Classical map associated with a representation and asymptotic results

In the toroidal representation, the compatibility condition $Q(V)\overline{\Lambda} = \overline{\boldsymbol{n}}$ is equivalent to the stationarity of the action for the projected norm at the origin of the relative phases. If the reduced Hessian is positive definite and the required regularity conditions hold, a local Gaussian approximation is obtained. Extending it to the full integral requires controlling the remaining contributions (Appendix H). The macroscopic analysis of the projected-state transitions leads to a classical map defined by the representation label Λ and the frame V (Appendix I):

$$C_{\Lambda,V,U}(\boldsymbol{\phi}) = Q(U t_{\boldsymbol{\phi}} V)\overline{\Lambda} = \operatorname{diag}\left[U t_{\boldsymbol{\phi}} \rho_{\Lambda}(V) t_{\boldsymbol{\phi}}^{\dagger} U^{\dagger}\right], \qquad \mathcal{R}_{\Lambda,V,U} = C_{\Lambda,V,U}(\mathbb{T}^{M-1}). \tag{4.6}$$

By the Schur–Horn theorem, $\mathcal{R}_{\Lambda,V,U} \subseteq \Pi(\overline{\Lambda})$, although the image does not necessarily coincide with the permutahedron.

For normalized final occupations at a uniformly positive distance from $\mathcal{R}_{\Lambda,V,U}$, Appendix I establishes an exponential suppression bound, provided that the initial projected norms are positive and not exponentially small. Unlike the Gaussian approximations, this bound does not require regularity of the map. Determining the prefactors and relative phases governing interference between stationary points, developing general semiclassical approximations for representation matrix elements of $D^{\Lambda}(U)$ beyond the fully symmetric case, and characterizing how these approximations relate to the conditional physical probabilities of the irreducible sectors remain open problems.

Supplementary material and reproducibility. The mathematical appendices supporting this proposal, together with the Wolfram Language code associated with the numerical calculations presented in this work, are available in the following GitHub repository.

5 Research problem and objectives

The preceding chapters presented the semiclassical correspondence between multiphoton probabilities and the measure induced by the classical map, the exact decomposition of these probabilities into irreducible sectors, and the constructions based on projected states and sectorial maps. These results provide a starting point for investigating how sector contributions relate to the multiphoton distribution and how classical maps can guide the development of semiclassical approximations for representation matrix elements.

5.1 Statement of the scientific problem

The classical map developed in our previous work [12] and the Schur–Weyl decomposition of Eq. (3.5) provide complementary descriptions of multiphoton scattering, but they do not yet determine how the contributions of individual irreducible sectors relate to the macroscopic probability distribution. Open questions concern the behavior near caustics and the exact suppression rates outside the classically allowed region, the role of populations and coherences within weight spaces, and semiclassical approximations for representation matrix elements beyond the fully symmetric representation. The central problem of this research is therefore to establish quantitative relations among the classical map and its induced measure, multiphoton probabilities with partial indistinguishability, and the contributions of the irreducible sectors of $U(M)$ in the many-photon limit, identifying the conditions under which semiclassical approximations can be obtained.

5.2 General objective

To establish quantitative correspondences among the scattering probabilities of partially indistinguishable photons in the many-photon limit $N \to \infty$ with a fixed number of modes M, the semiclassical behavior of irreducible representations of $U(M)$ in the limit of large Young diagrams, and classical multiport interferometry with partially coherent light, described by the classical map and its induced measure on the probability simplex.

5.3 Specific objectives

The specific objectives are organized around the three correspondences underlying this research.

1. Classical interferometry and multiphoton interference

To determine how the preimages and singularities of the classical map affect the asymptotic behavior of multiphoton probabilities, beginning with three-mode interferometers.

- **1.1.** To identify and classify the singularities of the classical map in selected configurations through its preimages and Jacobian, and quantify their effects on the probabilities.
- **1.2.** To determine bounds and decay rates for the probabilities in classically forbidden regions, establishing their relation to bounds based on relative entropy.
- **1.3.** To formulate and test uniform Airy-type approximations near fold caustics and WKB connection formulas between allowed and forbidden regions, under appropriate conditions.

2. Multiphoton interference and representation theory

To quantify how partial indistinguishability determines the contributions of the irreducible sectors of $U(M)$ and their weight spaces in the limit of large Young diagrams, beginning with $U(3)$.

- **2.1.** To characterize the probability distribution over irreducible representations of $U(M)$ and quantify the concentration of their normalized Young diagrams around the spectrum of the effective single-photon matrix, using Keyl–Werner bounds conditioned on fixed initial occupations.
- **2.2.** To determine how the populations and coherences of the physical density matrices $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$ enter the conditional probabilities and how these contribute to the full multiphoton distribution.
- **2.3.** To characterize how the choice of internal states favors particular irreducible sectors and affects populations and coherences within their weight spaces, and to extract information about the representation matrix elements of $D^{\Lambda}(U)$ from the resulting multiphoton probabilities.

3. Classical interferometry and representation theory

To construct semiclassical approximations for matrix elements of irreducible representations of $U(M)$ beyond the fully symmetric sector, using virtual states, multicoherent states and their projections onto weight spaces, compatible frames, and associated classical maps.

3.1. To characterize which directions within weight spaces can be described by projections of multicoherent states associated with compatible frames, and to formulate suitable integral representations for matrix elements of $D^{\Lambda}(U)$, deriving their stationarity conditions and their relation to the associated classical maps.

3.2. To determine, for selected families of $U(3)$ representations, the contributions of stationary points, their prefactors and relative phases, and to compare the resulting asymptotic formulas with exact and numerical results.

5.4 Scope and expected contributions

The research will initially focus on three-mode interferometers and representations of $U(3)$, where weight spaces with internal multiplicity $m_{\Lambda}(\boldsymbol{n}) > 1$ occur and the classical map may have interior caustics. Using Gelfand–Tsetlin bases, the isospin structures of $SU(3)$, and the exact and numerical results already obtained, quantitative correspondences will be evaluated in specific configurations, within the general framework of the limit $N \to \infty$ with fixed M.

For the first correspondence, the singularities of the classical map and their effects on multiphoton probabilities will be characterized, determining bounds and rates of exponential suppression, and formulating Airy- and WKB-type approximations in configurations where they apply. For the second, the contributions of irreducible sectors to the multiphoton distribution will be quantified, considering the concentration of their probability distributions and the populations and coherences of the physical density matrices $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$. The extent to which the choice of internal states favors particular sectors and affects these populations and coherences will be evaluated, alongside the information that the resulting multiphoton probabilities can yield about the representation matrix elements of $D^{\Lambda}(U)$.

Finally, for the third correspondence, semiclassical approximations for representation matrix elements beyond the fully symmetric case will be constructed, determining stationary contributions, prefactors, and relative phases through virtual states and constructions based on multicoherent states, their projections onto weight spaces, compatible frames, and associated classical maps. The analytical results will be compared with exact and numerical calculations, distinguishing physical sector probabilities from transitions of projected pure states. A general treatment of more highly degenerate singularities, complex stationary points, and the systematic extension to an arbitrary number of modes will remain possible directions for future work.

5.5 Methodology

The methodology will combine representation theory, semiclassical methods, and symbolic and numerical calculations, building on the exact results already developed. The research will initially focus on three-mode interferometers, considering sequences of occupations $\boldsymbol{n}$ and representations Λ such that $\boldsymbol{n}/N \to \overline{\boldsymbol{n}}$ and $\Lambda/N \to \overline{\Lambda}$ as $N \to \infty$, with fixed M.

For the first correspondence, the classical map $C_{\rho,U}$ will be analyzed through its preimages and Jacobians, identifying singularities, caustics, and voids. The induced classical measure will be compared with multiphoton probabilities for increasing values of N, evaluating contributions from different preimages and their relative phases where appropriate. Building on relative-entropy bounds, the rates of exponential suppression will be determined, and Airy- and WKB-type approximations will be formulated in configurations where the singularities and transitions between allowed and forbidden regions justify their use.

For the second correspondence, the exact decomposition of Eq. (3.5) and the concentration bounds associated with the Keyl–Werner theorem will be used. The probability distributions of the irreducible sectors will be calculated and the conditional physical probabilities will be analyzed using $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$ and its evolution under $D^{\Lambda}(U)$, evaluating how their populations and coherences contribute to the total distribution. For $SU(3)$, the physical isospin distributions and their moments will be determined. By varying the internal-state overlaps encoded in Γ, the extent to which specific irreducible sectors are favored and how populations and coherences change will be evaluated, alongside the information the

resulting multiphoton probabilities can yield about the representation matrix elements of $D^{\Lambda}(U)$.

Finally, for the third correspondence, constructions based on exterior powers, highest-weight orbits, and toroidal projections (Appendices C, H, and I) will be implemented, using the methods based on virtual single-photon states as a reference. For $U(3)$, compatible frames and projections of multicoherent states will be characterized through their isospin amplitudes, inner products, and Gram matrices to determine which directions in weight spaces they can describe. Using stationarity conditions, reduced Hessians, and classical maps associated with irreducible representations, semiclassical approximations for representation matrix elements beyond the fully symmetric case will be developed in selected configurations. The contributions of stationary points, their prefactors and relative phases will be determined, together with the conditions of validity and exponential suppression bounds. The results will be compared with exact and numerical calculations, distinguishing transitions of projected pure states from physical sector probabilities.

5.6 Ethical considerations

The research is theoretical and computational and therefore does not involve experiments with humans or animals or the processing of personal data. The main ethical considerations concern academic integrity and the reproducibility of the results. Proper attribution of previous work and documentation of the analytical and numerical procedures used will be strictly ensured. Furthermore, proved results, numerical evidence, and hypotheses yet to be verified will be explicitly distinguished, stating the conditions of validity and limitations of the approximations developed.

5.7 Timeline

The work plan covers doctoral semesters 7–10 and is organized around the three correspondences underlying the specific objectives. Building on the results already obtained, the analysis of the classical map, the characterization of sector contributions, and the development of semiclassical approximations for representation matrix elements will be carried out.

Table 2: Tentative timeline.

Period	**Planned activities and results**
Semester 7	Characterization of singularities, caustics, and voids of the classical map in three modes using preimages and Jacobians. Calculation of bounds and rates of exponential suppression and evaluation of Airy/WKB approximations in appropriate configurations, with comparisons against exact probabilities.
Semester 8	Calculation of the probability distributions of the irreducible sectors and their concentration. Analysis of conditional physical probabilities and the populations and coherences of $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$. Evaluation of how internal-state overlaps affect sector contributions and what information the resulting multiphoton probabilities can yield about the representation matrix elements of $D^{\Lambda}(U)$.
Semester 9	Characterization of internal multiplicity directions using compatible frames and projections of multicoherent states. Development of semiclassical approximations for representation matrix elements of $D^{\Lambda}(U)$ in selected $U(3)$ representations, using virtual states and toroidal projections. Determination of stationary contributions, prefactors, and relative phases. Quantification of their relation to sector probabilities and the induced classical measure, with exact and numerical comparisons.
Semester 10	Integration and validation of results, determination of the conditions of validity of the approximations, and identification of open problems. Preparation of manuscripts, final thesis writing, and defense.

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